

Konstantinos Christopher Tsiolis

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PROFILE

I am a third-year PhD student in Statistics at the University of Toronto and the Vector Institute, supervised by Prof. Murat Erdogdu. My research focuses on theoretical aspects of machine learning.

EDUCATION

Doctor of Philosophy (PhD), Statistics <i>University of Toronto</i> - Supervisor: Prof. Murat Erdogdu - Cumulative GPA: 4.0/4.0	2023 – Present Toronto, ON
Master of Science (MSc), Mathematics and Statistics <i>McGill University</i> - Supervisor: Prof. Adam Oberman - Cumulative GPA: 4.0/4.0 - Thesis Title: Contrastive Learning as Kernel Approximation	2022 – 2023 Montreal, QC
Bachelor of Science (BSc), Mathematics and Computer Science (Honours) <i>McGill University</i> Cumulative GPA: 3.99/4.0	2018 – 2021 Montreal, QC

PUBLICATIONS (* DENOTES EQUAL CONTRIBUTION)

- **Konstantinos Christopher Tsiolis**, Alireza Mousavi-Hosseini, and Murat A. Erdogdu. From Information to Generative Exponent: Learning Rate Induces Phase Transitions in SGD. In *Proceedings of the 39th International Conference on Neural Information Processing Systems (NeurIPS 2025)*. *To appear*.
- Jingyi He*, **KC Tsiolis***, Kian Kenyon-Dean*, and Jackie Chi Kit Cheung. Learning efficient task-specific meta-embeddings with word prisms. In *Proceedings of the 28th International Conference on Computational Linguistics*, pages 1229–1241. International Committee on Computational Linguistics, 2020.

RESEARCH EXPERIENCE

Research Assistant <i>Mila / McGill University Department of Mathematics and Statistics</i> - Supervisor: Prof. Adam Oberman - Developing, implementing, and testing novel approaches in knowledge distillation and self-supervised representation learning for computer vision	Summer 2021 Montreal, QC
Research Assistant <i>Mila / McGill Reasoning and Learning Lab</i> - Supervisor: Prof. Jackie Chi Kit Cheung - Natural language processing research on word embeddings and computational pragmatics - Presenting my work at reading groups and conferences	Summer 2019 and 2020 Montreal, QC

TEACHING EXPERIENCE

- Teaching Assistant**
- *At the University of Toronto (2023 – Present)*: STA2311 (Advanced Computational Methods for Statistics I), STA314 (Statistical Methods for Machine Learning I), STA261 (Probability and Statistics II), STA221 (The Practice of Statistics II), STA130 (An Introduction to Statistical Reasoning and Data Science)
 - *At McGill University (2021 – 2022)*: MATH 222 (Calculus 3), COMP 251 (Algorithms and Data Structures)
 - Prepared material for and conducted lectures, tutorials, exam review sessions, office hours, answering questions on online forums (e.g., Piazza), grading assignments and exams, and invigilation

LEADERSHIP AND SERVICE

- President of the Statistics Graduate Student Union (SGSU)** 2024 – 2025
University of Toronto Department of Statistical Sciences Toronto, ON
- Elected representative of the PhD and MSc students in my department
 - Organizing academic and social events throughout the year (e.g., Graduate Student Research Day, NSERC Award Application Workshop, Park Day, Game Nights)
- Conference Organizer and Reviewer**
- Member of the Local Organizing Committee for *The Fast and Curious 2: MCMC in Action* (2025)
 - Reviewer for the *Workshop on High Dimensional Learning Dynamics* (2023 – 2025)
- Academic Lecturer at the McGill Artificial Intelligence Society** 2020 – 2021
McGill University Montreal, QC
- Volunteer role as part of a student club
 - Giving lectures and writing a textbook for a student-run machine learning course
 - Holding office hours and one-on-one meetings with students

AWARDS

- NSERC Postgraduate Scholarship – Doctoral (PGS D)** 2024 – 2027
- Award value: 40,000 CAD per year
 - Awarded to outstanding Canadian students pursuing a PhD in the natural sciences or engineering
- Queen Elizabeth II Graduate Scholarship in Science and Technology (QEII-GSST)** 2023
- Awarded to outstanding students pursuing a PhD at an Ontario university
- Faculty of Arts and Science Top (FAST) Doctoral Fellowship** 2023 – 2027
- 4-year fellowship awarded to outstanding Canadian PhD students at the University of Toronto
- FRQNT Master’s (B1X) Research Scholarship** 2022
- Awarded to outstanding research MSc students in Quebec
 - Awarded in 2022, but funding deferred to 2023 due to NSERC award
- NSERC Canada Graduate Scholarships - Master’s Program (CGS M)** 2022 – 2023
- Awarded to outstanding research MSc students in Canada
- Wing Hing Chan Scholarship in Science** 2021
- Awarded to an outstanding McGill student in the Faculty of Science
- Alan Shaver Scholarship** 2021
- Awarded to an outstanding McGill student in the Faculty of Science
- IVADO Undergraduate Introduction to Research Scholarship** 2021
- Funding awarded to undergraduate students pursuing summer research in data science
- Anne Molson Scholarship** 2020
- Awarded to a McGill student in the Physical Sciences prior to their final year
- Emily Ross Crawford Scholarship** 2020
- Awarded to a McGill student in the Faculty of Science based on academic achievement in the first two years
- NSERC Undergraduate Student Research Award** 2020
- Funding awarded to undergraduate students pursuing summer research in natural sciences or engineering
- John Mahon Scholarship** 2018 – 2021
- Annually renewable entrance scholarship at McGill University